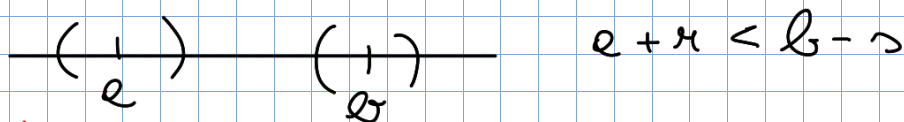


$\mathbb{R}$ ha le seguenti proprietà di separazione:

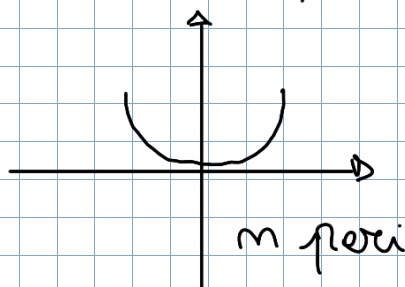
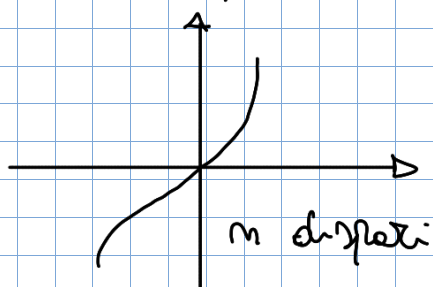
$a, b \in \mathbb{R}$ con $a \neq b \quad \exists r, s > 0 \quad B(a, r) \cap B(b, s) = \emptyset$



Funzioni elementari:

1) Funzione potenza

$m \in \mathbb{N} \quad f(x) = x^m \quad \forall x \in]-\infty; +\infty[$

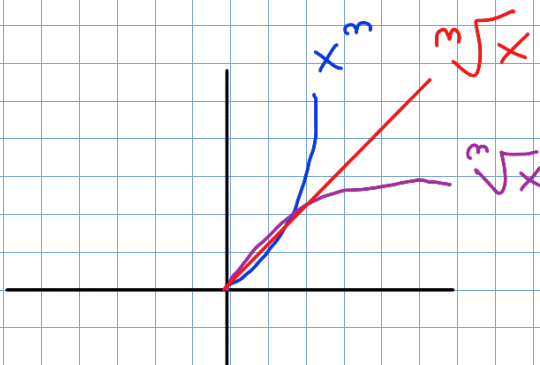
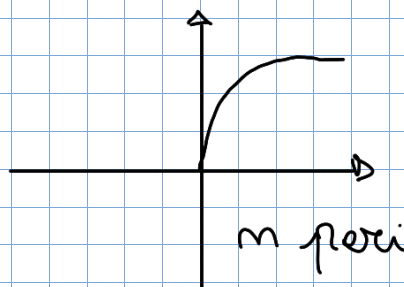
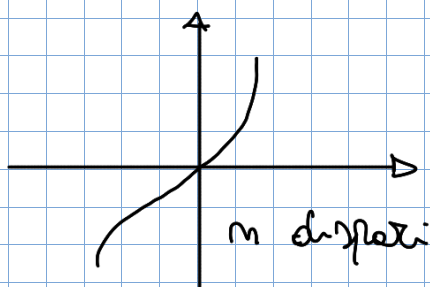


m dispari $\Rightarrow$ invertibile in $]-\infty; +\infty[$

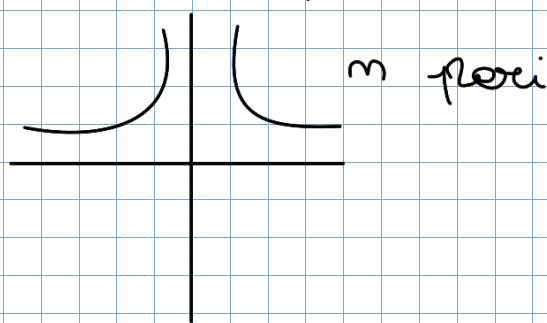
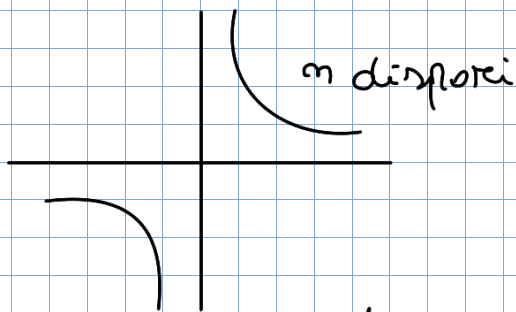
m pari $\Rightarrow$ " " in $[0, +\infty[$ e $]-\infty, 0]$

inversa: $\sqrt[m]{x}$

$f(x) = \sqrt[m]{x} \quad \forall x \geq 0$ se m pari
 $\forall x \in \mathbb{R}$ se m dispari



$$m \in \mathbb{N} \quad f(m) = x^{-m} \quad \forall x \neq 0 \quad x^{-m} = \frac{1}{x^m}$$



$$m \in \mathbb{N} \quad f(x) = x^{\frac{1}{m}} \quad (= \sqrt[m]{x} \quad \text{se } x \geq 0)$$

$$r = \frac{p}{q} \in \mathbb{Q} \quad f(x) = x^{\frac{p}{q}} = (\sqrt[q]{x})^p$$

$$\lambda \in \mathbb{R} - \mathbb{Q} \quad f(x) = x^\lambda \quad \forall x \geq 0 \quad \text{se } \lambda > 0$$

$$\forall x > 0 \quad \text{se } \lambda < 0$$

x^π ($\pi \in \mathbb{Q}$), x^λ ($\lambda \in \mathbb{R} - \mathbb{Q}$) all'infinito e $x^{\frac{1}{\pi}}$, $x^{\frac{1}{\lambda}}$ sono strettamente crescenti in $]0, +\infty[$

2) Polinomi

$$f(x) = a_0 x^m + a_1 x^{m-1} + \dots + a_{m-1} x + a_m$$

$$a_0, \dots, a_m \in \mathbb{R} \quad m \in \mathbb{N}_0 \quad a_1, \dots, a_m \text{ coefficienti}$$

$$\forall x \in \mathbb{R}$$

$$m=0 \text{ polinomio costante } f(x) = a_0 \quad \forall x \in \mathbb{R}$$

$$\text{se } a_0 = 0 \text{ polinomio nullo}$$

$$f(x) = a_0 x^m + \dots + a_m$$

$$g(x) = b_0 x^r + \dots + b_r$$

sono uguali (cioè $f(x) = g(x)$ $\forall x$) se e solo se sono due polinomi identici cioè $m = r$ $\rightarrow$ grado

(principio di identità del polinomio)

Dati 2 polinomi $f, g \in \mathbb{Q}(x), r(x)$ con grado di $r < \text{grado di } g$ tali che $f(x) = g(x)q(x) + r(x)$

teorema di Ruffini

il polinomio $f(x)$ è divisibile per $x - c \Leftrightarrow f(c) = 0$

3) Funzione razionale fratta

$$f(x) = \frac{a_0 x^n + \dots + a_m}{b_0 x^p + \dots + b_r}$$

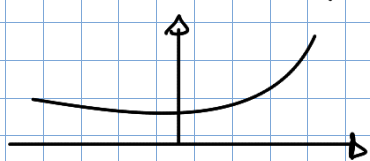
$$\begin{aligned} a_0, \dots, a_m, b_0, \dots, b_r &\in \mathbb{R} \\ n &\in \mathbb{N}_0 \\ p &\in \mathbb{N} \end{aligned}$$

f è definita in $\mathbb{R} \setminus \{c \in \mathbb{R} : b_0 c^p + \dots + b_r = 0\}$

4) Funzione esponenziale

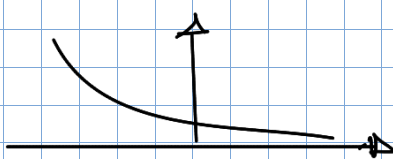
$$a \in \mathbb{R} \quad a > 0, a \neq 1 \quad f(x) = a^x \quad \forall x \in \mathbb{R}$$

$$a^x > 0 \quad \forall x \in \mathbb{R}$$



$$a > 1$$

strett crescente



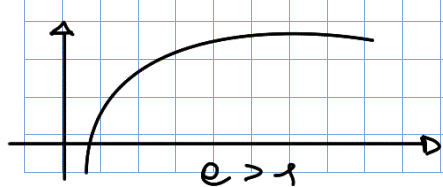
$$0 < a < 1$$

strett decrescente

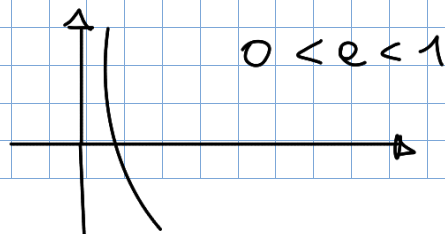
$$f:]-\infty; +\infty[\rightarrow]0; +\infty[$$

Funzione logaritmica

$$f^{-1}:]0; +\infty[\rightarrow]-\infty; +\infty[$$

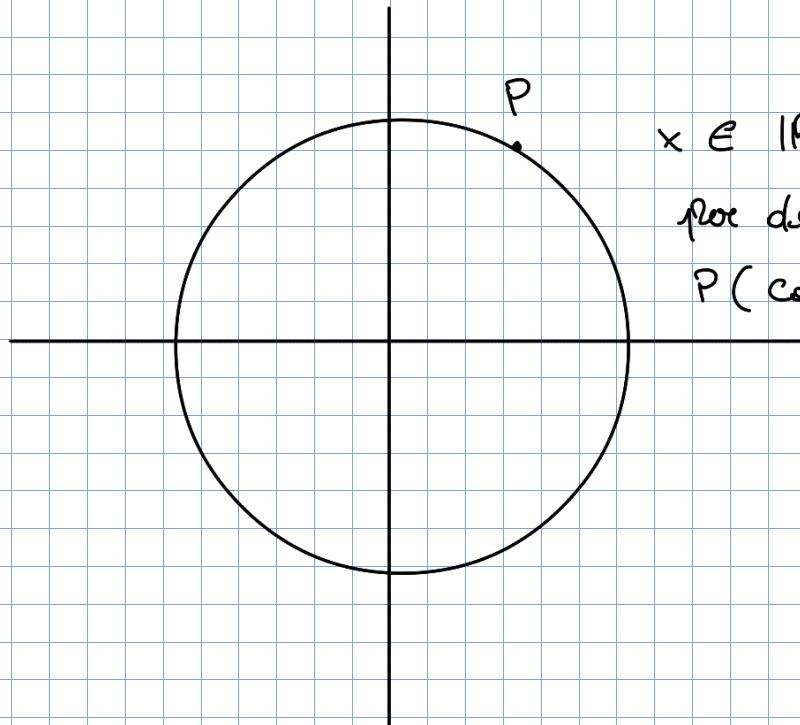
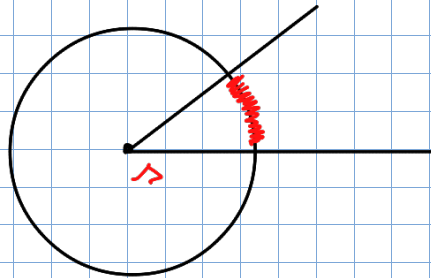
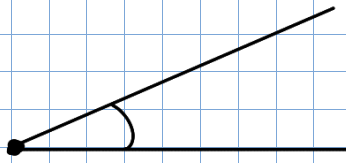


f^{-1} è la funzione logaritmo



$$f^{-1}(f(x)) = x \quad f(f^{-1}(y)) = y$$

5) funzioni trigonometriche



$$x \in \mathbb{R}$$

per definizione

$$P(\cos x, \sin x)$$

Se $x \in \mathbb{R}$, $\cos x$ ($\sin x$) è l'ascissa (l'ordinata) del secondo estremo di un arco della circonferenza trigonometrica avente primo estremo $A(1,0)$ e misura in radianti x .

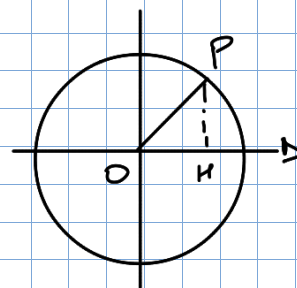
$$-1 \leq \cos x \leq 1$$

$$-1 \leq \sin x \leq 1$$

$$\cos^2 x + \sin^2 x = 1$$

$$\forall x \in \mathbb{R}$$

$$\forall x \in \mathbb{R}$$

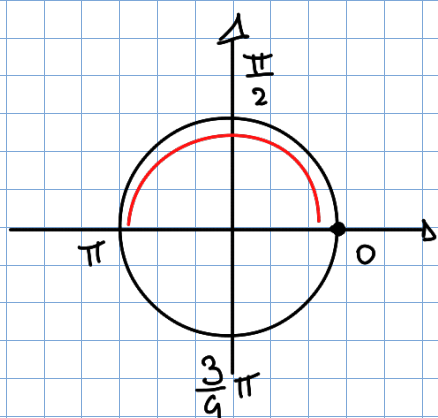


$$PH = \sin x$$

$$OH = \cos x$$

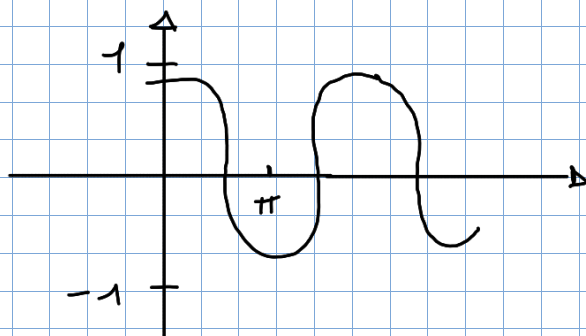
$$OP = 1$$

$$\left. \begin{aligned} \cos(x + 2k\pi) &= \cos x \\ \sin(\quad) &= \sin x \end{aligned} \right\} \forall x \in \mathbb{R}, \forall k \in \mathbb{R}$$



$\cos 0 = 1$ in $[0, \pi]$ $f(x) = \cos x$ è
 $\cos \pi = -1$ strett. decr. e assume
 tutti i valori tra -1 e 1 quindi è
 invertibile e abbiamo
 $\cos: [0, \pi] \rightarrow [-1, 1]$
 $\arccos: [-1, 1] \rightarrow [0, \pi]$

$\forall d \in [-1, 1]$ def $\arccos d =$ l'unico $x \in [0, \pi]: \cos x = d$



$\cos x = d$

$\vee d \in [-1, 1] \quad \text{sol} \pm \arccos d + 2k\pi$

$\vee d > 1$ oppure $d < -1$ nessun sol.

$\cos x > d$

$\vee -1 \leq d < 1$

$\text{sol: } -\arccos d + 2k\pi < x < \arccos d + 2k\pi$

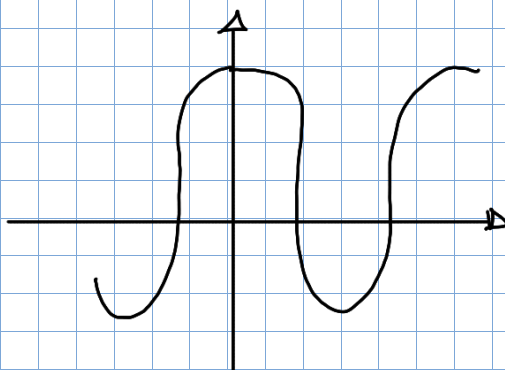
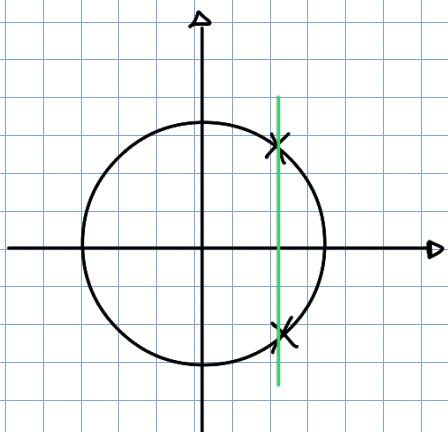
$\vee d \geq 1$

nessuna soluzione

$\vee d \leq -1$

$\forall x \in \mathbb{R}$ è sol

$\cos x < d$



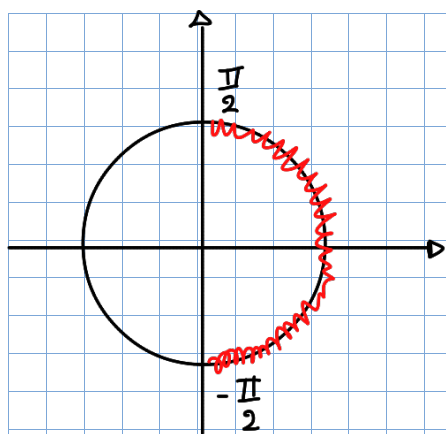
$\vee -1 < d < 1 \quad \arccos d < m < 2\pi - \arccos d$

$\vee d \leq -1$

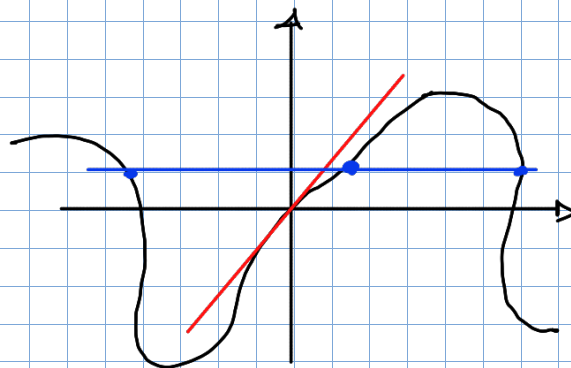
nessuna soluzione

$\vee d > 1$

$\forall x \in \mathbb{R}$ è sol



$$\sin(-x) = -\sin x \quad \forall x \quad (\text{funzione dispari})$$



$$\sin\left(-\frac{\pi}{2}\right) = -1$$

$$\sin\left(\frac{\pi}{2}\right) = 1$$

in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ è stretto crescente e assume tutti i valori fra -1 e 1

$$\sin: \left[\frac{\pi}{2}, -\frac{\pi}{2}\right] \rightarrow [-1, 1]$$

$$\arcsin: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\sin x = d \quad \text{se } -1 \leq d \leq 1$$

$$\text{sol } x = \arcsin d + 2k\pi$$

$$\pi - \arcsin d + 2k\pi$$

$$\sin x < d$$

$$-\pi - \arcsin d + 2k\pi < x < \arcsin d + 2k\pi \quad \text{se } -1 < d \leq 1$$

$$\forall x \in \mathbb{R} \quad \text{e' sol se } d > 1$$

$$\text{non sol.} \quad \text{se } d \leq -1$$

$$-\pi - \arcsin d + 2k\pi < x < \pi - \arcsin d + 2k\pi = \quad \text{se } -1 < d < 1$$

$$\text{non sol} \quad \text{se } d \geq 1$$

$$\forall x \in \mathbb{R} \quad \text{e' sol} \quad \text{se } d < -1$$

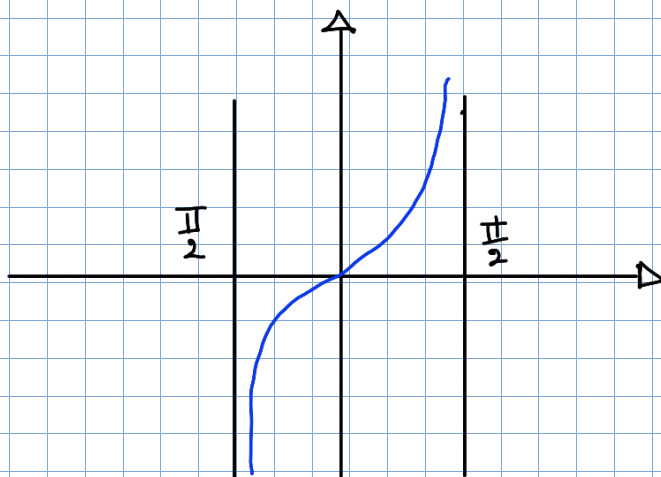
Formule di Brachione (Che cosa sono?)

Definizione tangente

$$\tan x = \frac{\sin x}{\cos x} \quad x \neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$$

$$\tan(x + k\pi) = \tan x \quad \forall x, \quad \forall k \in \mathbb{Z}$$

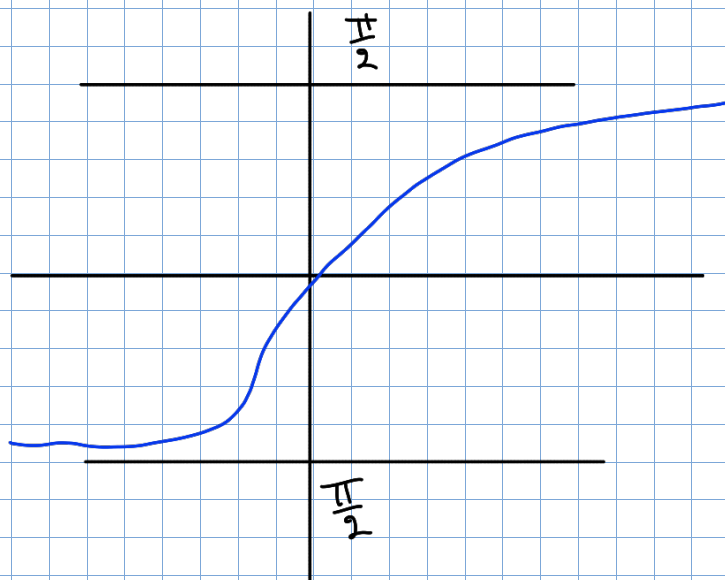
$$\tan(-x) = \frac{\sin(-x)}{\cos(-x)} = \frac{-\sin x}{\cos x} = -\tan x$$



in $]-\frac{\pi}{2}, \frac{\pi}{2}[$
 $f(x) = \tan x$
 è strettamente
 crescente e assume tutti
 i valori reali

$$\tan:]-\frac{\pi}{2}; \frac{\pi}{2}[\rightarrow]-\infty; +\infty[$$

$$\arctan:]-\infty; +\infty[\rightarrow]-\frac{\pi}{2}, \frac{\pi}{2}[$$



$$\tan x = d$$

$$\tan x > d$$

$$\tan x < d$$

$$x = \arctan d + k\pi \quad \forall d \in \mathbb{R}$$

$$\arctan d + k\pi < x < \frac{\pi}{2} + k\pi$$

$$-\frac{\pi}{2} + k\pi < x < \arctan d + k\pi$$

$$f(x) = \sqrt{\arctan x - \frac{x}{x^2-1}} \quad \text{e' funz. elementare}$$

Esercizi: sull'insieme di definizione

$$f(x) = 2 \frac{\sqrt{x-1}}{|x|-3}$$

$$\begin{cases} x-1 \geq 0 \\ |x|-3 \neq 0 \end{cases}$$

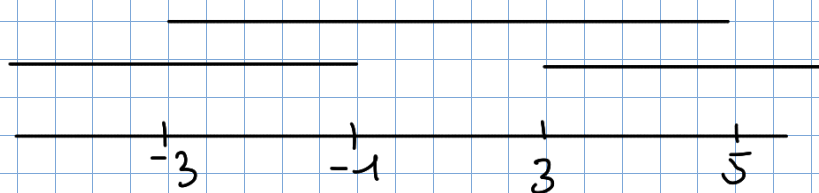
$$\begin{cases} x \geq 1 \\ x \neq 3; x \neq -3 \end{cases}$$

$$f: ([1, 3[\cup]3, +\infty[) \rightarrow \mathbb{R}$$

$$f(x) = \arcsin(|x-1|-3)$$

$$-1 \leq |x-1|-3 \leq 1 \rightarrow 2 \leq |x-1| \leq 4$$

$$\begin{cases} |x-1| \geq 2 \\ |x-1| \leq 4 \end{cases} \Rightarrow \begin{cases} x-1 \leq -2 \vee x-1 \geq 2 \\ -4 \leq x-1 \leq 4 \end{cases} \Rightarrow \begin{cases} x \leq -1 \vee x \geq 3 \\ -3 \leq x \leq 5 \end{cases}$$



$$f: [-3, -1] \cup [3, 5] \rightarrow \mathbb{R}$$

$$f(x) = \sqrt[4]{\sqrt[3]{x-2}-1}$$

$$\sqrt[3]{x-2}-1 \geq 0 \rightarrow \sqrt[3]{x-2} \geq 1 \rightarrow x-2 \geq 1 \rightarrow x \geq 3$$

$$f: [3; +\infty[\rightarrow \mathbb{R}$$

$$e^x$$

$$\log x = \log_e x$$

$$e > 1$$

$$f(x) = \log \sqrt{\frac{|x|-2}{x^2+3}}$$

$$\begin{cases} \sqrt{\frac{|x|-2}{x^2+3}} > 0 \\ \frac{|x|-2}{x^2+3} \geq 0 \\ x^2+3 \neq 0 \end{cases} \Rightarrow \begin{cases} |x|-2 > 0 \\ \Downarrow \\ x > 2 \vee x < -2 \end{cases}$$

$$f: (-\infty, -2] \cup [2, +\infty) \rightarrow \mathbb{R}$$

Numeri Complessi

$$C = \{(a, b) : a, b \in \mathbb{R}\} \quad \text{coppie ordinate}$$

$$(0, 0) = 0 \quad \text{zero}$$

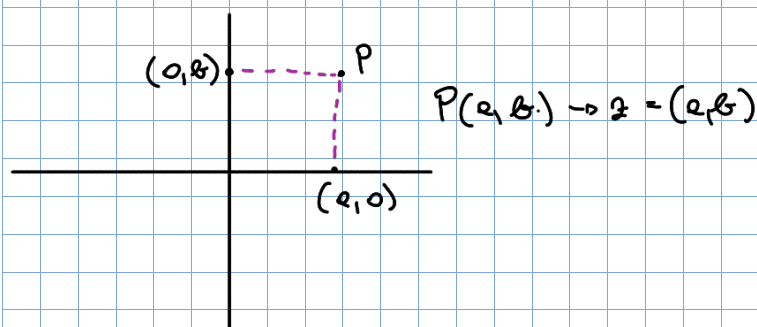
$$z = (a, b) \quad -z = (-a, -b) \quad \text{opposto}$$

$$\bar{z} = (a, -b) \quad \text{coniugato}$$

le nostre nuove x

$$(a, 0) \longrightarrow \text{numero complesso reale}$$

$$(0, b) \longrightarrow \text{numero immaginario puro}$$



2 numeri Complessi z, w sono uguali se:

Definizione: $z = w$ se $a = c, b = d$

se $z \neq w$ non c'è un ordine

$$\begin{aligned} z &= (a, b) \\ w &= (c, d) \end{aligned}$$

$$i = (0, 1) \quad \text{unità immaginaria}$$

$$1 = (1, 0) \quad \text{unità reale}$$

Definizione somma $\rightarrow$ vale la prop. commutativa

$$z + w = (a + c, b + d)$$

$$z = (a, b)$$

$$w = (c, d)$$

$$z + 0 = 0$$

$$z + \bar{z} = (a + a, b - b) = (2a, 0) \text{ è un numero complesso reale}$$

Definizione moltiplicazione

$$z \cdot w = (ac - bd, ad + bc)$$

$$z \bar{z} = (a, b)(a, -b) = (a^2 + b^2, 0) \text{ è un numero complesso reale}$$

$$z = (a, b) \quad |z| = \sqrt{a^2 + b^2} \quad \text{modulo di } z$$

$$\text{se } z = (a, 0) \quad |z| = \sqrt{a^2 + 0} = |a|$$

Dimostrare che ponendo $a \rightarrow (a, 0)$ si ottiene una corrispondenza biunivoca fra $\mathbb{R}$ e l'insieme complessi reali che conserva le operazioni

quindi si può identificare $\mathbb{R} \subseteq \mathbb{C}$ e $a = (a, 0)$ identificazione

$$\begin{array}{ccc} 2 & + & 4 = 6 \\ \downarrow & & \downarrow \\ (2, 0) & + & (4, 0) = (6, 0) \end{array}$$

$$\begin{array}{ccc} 2 & \cdot & 4 = 8 \\ \downarrow & & \downarrow \\ (2, 0) & \cdot & (4, 0) = (8, 0) \end{array}$$

$$\text{"} \\ (2 \cdot 4 - 0 \cdot 0, 2 \cdot 0 + 0 \cdot 4) = (8, 0)$$

$$z \cdot \bar{z} = |z|^2 = |\bar{z}|^2 = |-z|^2$$